计算机科学与技术学院神经网络与深度学习课程实验报告

实验题目：极市平台新手任务实践		学号：202200130077
日期：2024.11.18	班级：22 级人工智能班	姓名：薛润泽
Email：		

实验目的：

通过完成新手任务实践，了解使用极市平台，熟悉实验流程以及未来实验平台。

实验结果：（截图本人测试任务成功页，如下图）



实验步骤：（选择训练、测试中重要的步骤截图）：

训练成功：



训练过程日志（成功训练完 299 个数据）：

cvmart.net/log/161009?flag=train

```
1543                 all        429       2061      0.911      0.818      0.863      0.556
1544
1545       Epoch    GPU_mem   box_loss   obj_loss   cls_loss  Instances       Size
1546     298/299       4.7G    0.02091    0.01409   0.001928        208    298/299       4.7G    298/299       4.7G      0.021    0.01414   0.001965         86        640: 100%|
         ██████████| 161/161 [01:44<00:00,  1.54it/s]
1547                Class     Images  Instances          P          R      mAP@.5 mAP@.5:.95: 100%|██████████| 14/14 [00:09<00:00,  1.44it/s]
1548                  all        429       2061       0.92      0.815      0.862      0.556
1549
1550       Epoch    GPU_mem   box_loss   obj_loss   cls_loss  Instances       Size
1551     299/299       4.7G    0.02117    0.01455   0.002097        106    299/299       4.7G    299/299       4.7G    0.02119    0.01443   0.002033         109        640: 10
         0%|          | 161/161 [01:46<00:00,  1.51it/s]
1552                Class     Images  Instances          P          R      mAP@.5 mAP@.5:.95: 100%|██████████| 14/14 [00:09<00:00,  1.42it/s]
1553                  all        429       2061      0.918      0.815      0.862      0.557
1554
1555   300 epochs completed in 10.752 hours.
1556   Optimizer stripped from /project/train/models/train/exp2/weights/last.pt, 14.4MB
1557   Optimizer stripped from /project/train/models/train/exp2/weights/best.pt, 14.4MB
1558
1559   Validating /project/train/models/train/exp2/weights/best.pt...
1560   Fusing layers...
1561   Model summary: 213 layers, 7018216 parameters, 0 gradients, 15.8 GFLOPs
1562                Class     Images  Instances          P          R      mAP@.5 mAP@.5:.95: 100%|██████████| 14/14 [00:05<00:00,  2.68it/s]
1563                  all        429       2061      0.912      0.801      0.857      0.558
1564               person        429       1032      0.932      0.897      0.942      0.675
1565                  hat        429        863      0.917      0.849      0.887      0.548
1566                 head        429        166      0.886      0.656      0.743      0.452
1567   Results saved to [1m/project/train/models/train/exp2[0m
1568   {"code":20000,"message":"执行成功","data":"成功"}
1569   2024-11-19 09:15:45.879 - {
1570     "train_log_path": {
1571       "name": "",
1572       "size": ""
1573     },
1574     "model_list": [
1575       {
```

测试日志：

cvmart.net/log/188931?flag=test

```
1   2024-11-19 14:05:56,781 - root - INFO - ================================================
2   2024-11-19 14:05:56,781 - root - INFO - log_path                        : None
3   2024-11-19 14:05:56,781 - root - INFO - log_level                       : None
4   2024-11-19 14:05:56,782 - root - INFO - config_params                   : {"algorithm_type":"object_detection_single_label","dataset_params":{"class_name
    s":["person","head","hat"],"dataset_root":"/home/data/","sample_path":"/home/sample/"},"enable_custom_metric":false,"model_data_path":"model_data","e
    nable_fps_metric":true,"enable_resource_metric":true,"fps_metric_params":{"weight":0.3,"benchmark":100},"metric_config":{"metric_name":"f-score","param
    s":{"beta":1,"iou_thresh":0.7,"use_macro":false}},"enable_data_template":false,"resource_metric_params":{"query_cpu":true,"query_gpu":true}}
5   2024-11-19 14:05:56,782 - root - INFO - message_save_path               : /usr/local/auto_test_client/test/188931/a/result.json
6   2024-11-19 14:05:56,782 - root - INFO - sample_visual_savepath          : /test-result/sample
7   2024-11-19 14:05:56,782 - root - INFO - test_task_id                    : 188931
8   2024-11-19 14:05:56,782 - root - INFO - callback_url                    : https://www.cvmart.net/cvmart-trains/api/ft/online-train/callback/service/a/188931
9   2024-11-19 14:05:56,782 - root - INFO - device                          : nvidia
10  2024-11-19 14:05:56,782 - root - INFO - use_algo_python_api             : True
11  2024-11-19 14:05:56,782 - root - INFO - show_result_gt                  : True
12  2024-11-19 14:05:56,782 - root - INFO - draw_fps                        : False
13  2024-11-19 14:05:56,782 - root - INFO - draw_prediction_label           : True
14  2024-11-19 14:05:56,782 - root - INFO - generate_wrong_result           : False
15  2024-11-19 14:05:56,782 - root - INFO - wrong_result_add_watermark      : True
16  2024-11-19 14:05:56,782 - root - INFO - wrong_result_save_path          : /test-result/error
17  2024-11-19 14:05:56,782 - root - INFO - wrong_result_save_quota         : 0
18  2024-11-19 14:05:56,782 - root - INFO - dynamic_mode                    : False
19  2024-11-19 14:05:56,782 - root - INFO - dynamic_id                      : None
20  2024-11-19 14:05:56,782 - root - INFO - dynamic_code_dir                : None
21  2024-11-19 14:05:56,782 - root - INFO - ================================================
22  2024-11-19 14:05:56,782 - root - INFO - config_params: {'algorithm_type': 'object_detection_single_label', 'dataset_params': {'class_names': ['person', 'head',
    'hat'], 'dataset_root': '/home/data/', 'sample_path': '/home/sample/'}, 'enable_custom_metric': False, 'model_data_path': 'model_data', 'enable_fps_metric': Tr
    ue, 'enable_resource_metric': True, 'fps_metric_params': {'weight': 0.3, 'benchmark': 100}, 'metric_config': {'metric_name': 'f-score', 'params': {'beta': 1, 'iou_thr
    esh': 0.7, 'use_macro': False}}, 'enable_data_template': False, 'resource_metric_params': {'query_cpu': True, 'query_gpu': True}}
23  2024-11-19 14:05:56,782 - root - INFO - using built-in plugins: algorithm_type=object_detection_single_label, is_custom_parser=False
24  2024-11-19 14:05:58,385 - root - INFO - DataGenerator `ObjectDetectionDataGenerator` init...
25  2024-11-19 14:05:58,386 - root - INFO - Using algorithm python API
26  2024-11-19 14:06:00,069 - yolov5 - INFO - YOLOv5 🚀 2024-11-19 Python-3.7.16 torch-1.11.0+cu113 CUDA:0 (Tesla T4, 14966MiB)
27
28  YOLOv5 🚀 2024-11-19 Python-3.7.16 torch-1.11.0+cu113 CUDA:0 (Tesla T4, 14966MiB)
```

结论分析与体会：

成功完成新手任务实验

平台使用反馈：（列举使用极市平台完成该任务遇到的困难或者麻烦，提出建设性意见）

意见反馈：
极市平台在分配资源时会根据 IP 地址来分配资源，这就导致同学们一起使用平台时造成挤兑现象。资源少，大家一起用，不够分配，导致大家都分配不到。
比如会出现：



就实验过程中遇到和出现的问题（算法相关或者是平台相关都可以），你是如何解决和处理的，自拟 1—3 道问答题：
问题：GitHub 中的资源，比如 requirements 列表是最新的 requirements。使用这个列表是无法与其他环境配对的，必须使用助教提供的 requirements（旧版）
解决方式：经过同学帮助，使用旧版的 requirements 就解决了环境问题